Arabesque

Element	Constraints
Supporting Leg	Rotated (sole of shoe at least partially showing) - Challenge: analyzing 3d image from 2d perspective Knee straight - Hip-knee-angle close to 180 degrees - Slightly past 180 degrees ok for hyperextension
Working Leg	Knee-angle-toe angle acute (slightly winged foot, penalize over-winging too!) Rotated, heel not visible from underrotation or overrotation (opening working leg hip may cause medial heel to appear, not enough rotation may cause lateral heel to appear)
Hips	Square Should not be opening hip of working leg
Chest & Back	Lifted and supported Sternum to center of head should be vertical line relative to rest of arabesque also need to make sure back is square - based on distance between right shoulder and left shoulder - they should be as close to each other as possible, mediapipe may not let them overlap but they should not be far apart
Head	Looking slightly upward over the finger Center of head should still be in line with sternum
Arm in Front of Body for 1st Arabesque	30-45 degree angle angle from shoulder to top of hand Angle from arm-wrist-fingers on arm in front should not be 180 degrees - Maybe 150 degrees for break in wrist Slight break in elbow too, ankle slightly less than 180 degrees Fingers soft (unsure how to evaluate)

Arm to Side of Body	Below shoulder Shoulder-elbow-wrist angle should be 180 degrees Fingers soft (unsure how to evaluate)

I think this project demonstrates how complicated and exact ballet positions are (even those that seemingly looks simple). Unveils how difficult ballet really is. You have so much to think about as a dancer and all of these different elements must be precise, perfectly placed, and coordinated (which is better demonstrated with dynamic movement over static poses).

- Created pose estimation landmarks
- Now i suppose we evaluate each of these technique criteria independently and generate an output if some are not met (output can depend on what the specific image has. For example, if the angle is significantly lower than needed, we output something specific…). Is there any more mathematically sophisticated way to do things or is this what an experienced quant trader / ml researcher would do?
- **Landmarks** = raw body points from the pose estimation model (we used mediapipe)
    - Stored as coordinates (e.g. (x, y, confidence))
- **Features** = values you compute from the landmarks
    - Examples
        - supporting_knee_angle = angle(hip, knee, ankle)
        - hip_tilt = difference between left hip height and right hip height
        - working_leg_height = ankle height relative to hip
        - front_arm_angle = angle from shoulder to wrist
    - They represent the measurable properties that body parts give rise to
    - **Interpretable features** = when humans can understand what the features mean
        - E.g. this feature measures knee straightness, therefore the feature is an interprtable feature
- **Criteria scores**
    - Based on the features, how well each technique criteria is satisfied
- **Feedback**
    - Use the criteria scores/evaluation to generate feedback for the dancer
- Pipeline: landmarks → interpretable features → criteria scores → feedback

Example
Image:
arabesque photo

Landmarks:
supporting hip, supporting knee, supporting ankle (specific body points)

Interpretable feature:
supporting knee angle = 168 degrees

Criterion score:
supporting leg straightness = moderate issue

Feedback:
"Lengthen the supporting knee; it appears slightly bent."

Step 1: identify landmarks with pose estimation model (mediapipe) (done)
Step 2: extract interpretable features from landmarks

See what each each landmark in mediapipe represents
https://developers.google.com/edge/mediapipe/solutions/vision/pose_landmarker

```
0 - nose
1 - left eye (inner)
2 - left eye
3 - left eye (outer)
4 - right eye (inner)
5 - right eye
6 - right eye (outer)
7 - left ear
8 - right ear
9 - mouth (left)
10 - mouth (right)
11 - left shoulder
12 - right shoulder
13 - left elbow
14 - right elbow
15 - left wrist
16 - right wrist
17 - left pinky
18 - right pinky
19 - left index
20 - right index
21 - left thumb
22 - right thumb
23 - left hip
24 - right hip
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```
25 - left knee
26 - right knee
27 - left ankle
28 - right ankle
29 - left heel
30 - right heel
31 - left foot index
32 - right foot index
```